

AIRA JOSH C. YNTE

Computer Engineer

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PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering graduate with hands-on experience in software development, embedded systems, and full-stack web applications. Passionate about designing robust systems, bridging the gap between hardware and software, and building performant user-facing applications. Skilled in React, TypeScript, C/C++, Flutter, and system architecture.

EXPERIENCE

- IC Layout Trainee** June 2026 – Present
- Xinyx Design Consultancy and Services Inc. / Xinyx Labs*
- Undergoing specialized training in IC Layout design, floorplanning, physical design verification (DRC/LVS), and parasitic extraction.
 - Gaining hands-on experience mapping logic layouts and verifying timing constraints in physical layout workflows.
- IC Design Intern (Design & Verification)** June 2025 – August 2025
- Xinyx Design Consultancy and Services Inc.*
- Designed and verified AMBA APB3 peripheral interfaces using SystemVerilog, achieving 100% code coverage across 20+ simulation test cases.
 - Analyzed waveforms and log profiles in behavioral simulations to verify timing constraints and protocol compliance.

EDUCATION

- Bachelor of Science in Computer Engineering (Magna Cum Laude)** 2022 – July 2026
- Colegio de Muntinlupa — Muntinlupa City, Metro Manila, Philippines*
- Specializing in Software Architecture & System Design; coursework in database design, computer networks, DSP, and system simulation.
 - Awarded Best Thesis for capstone project "AeroVit" (Gamified AI Fitness Platform combining embedded smartwatch biometrics & MediaPipe pose estimation).

KEY PROJECTS

- AeroVit — Gamified AI Fitness Platform (Thesis Project)** **Awarded Best Thesis** 2024 – 2026
- Developed an adaptive AI fitness system with ESP32-S3 FreeRTOS smartwatch firmware to track and stream heart rate and SpO2 biometrics over BLE.
 - Built a Flutter mobile app utilizing MediaPipe BlazePose (33 landmarks) for real-time posture tracking and AI form coaching.
 - Coded Flame 2D game and Solidity smart contracts on Sepolia supporting ERC-20 withdrawals with 5% burn.
- Fehuvia — Open Finance B2B SME Treasury Workstation** 2026
- Developed a B2B treasury workstation to streamline SME cash flow management, reducing obligation processing times by 40%.
 - Integrated secure Web3 transaction signing to automate and accelerate business settlements to under 2 seconds.
 - Engineered a robust backend to synchronize on-chain transaction logs, maintaining 99.9% financial data consistency.
- StickOut — Interactive VLSI EDA Editor** 2026
- Created HTML5 Canvas EDA viewport with coordinate transforms and grid snapping, rendering layouts at 60 FPS.
 - Coded same-layer wire auto-bridging algorithm to resolve crossovers and short-circuits in under 5ms.
 - Built double-pass regex parser and canvas rendering engine to render LaTeX subscripts (e.g. V_{DD}) in real-time.

SKILLS & COMPETENCIES

- Software Development:** Clean architecture, API engineering, full-stack debugging, performance optimization, modular system design
- System Design & Architecture:** Modular architecture planning, database schema design & planning, client-server models, decentralized system protocols
- Embedded & Hardware Engineering:** IC Layout design & physical verification (DRC/LVS), SystemVerilog logic verification (AMBA APB3), RTOS firmware, PCB layout & bring-up
- Design & Interactivity:** UI/UX wireframing (Figma), high-fidelity HTML5 canvas viewports, micro-interactions, motion choreography
- Technical Stack:** TypeScript, JavaScript, C/C++, Python, SQL (PostgreSQL, MySQL, SQLite), React, Next.js, Express, Node.js, Flutter, Tailwind CSS, Solidity, SystemVerilog, Git, Docker, KiCad, Firebase, MongoDB

CERTIFICATIONS & TRAINING

AI Fluency (Anthropic) · Model Context Protocol: Foundations & Advanced (Anthropic) · The AI Engineer Path (Scrimba) · Blockchain4Youth (Bitget) · On-the-Job Training Certificate (Xinyx Design) · Learn React & TailwindCSS (Scrimba) · TINA Design Suite (Hytec Power) · Understanding EDR (Cybersecurity) · Introduction & Intermediate Git (DataCamp)